ResearchForge
AI Training Operating System
Master Product Specification

1. Executive Vision

ResearchForge is a CLI-first operating system that transforms a research idea into a reproducible AI training pipeline. Instead of relying on a single LLM to generate notebooks and code, the platform orchestrates deterministic systems, search APIs, knowledge graphs, validation engines, and AI reasoning agents.

2. Problem Statement

Training modern ML and LLM systems requires stitching together dozens of disconnected tools for search, datasets, cleaning, experimentation, GPU provisioning, tracking, evaluation, and deployment. Existing platforms solve isolated problems rather than the complete workflow.

3. Product Principles

• Evidence before reasoning
• Deterministic execution before generation
• Reproducibility by default
• Explain every recommendation
• Every decision linked to evidence
• CLI-first, API-first, cloud optional

4. Target Users

ML engineers, AI researchers, startups, university labs, open-source contributors, enterprises building domain-specific models.

5. Competitive Landscape

Existing products: Together AI, Predibase, Modal, RunPod, Weights & Biases, Hugging Face, Axolotl, torchtune, LLaMA Factory, Lightning AI.
Gap: No unified operating system manages research, datasets, planning, execution, optimization, and reporting end-to-end.

6. High-Level System Architecture

CLI
↓
Intent Parser (Gemini)
↓
Parallel Evidence Search
↓
Evidence Store
↓
Knowledge Graph (Neo4j)
↓
Reasoning Layer
↓

Dataset Validation
↓
Training Planner
↓
Config Generator
↓
Execution Engine
↓
Experiment Manager
↓
Evaluation Engine
↓
AutoResearch Agent
↓
Reports & Deployment

7. Intent Parsing

Extract structured JSON: objective, task, modality, labels, evaluation metric, constraints, GPU availability, framework preference, expected output.

8. Evidence Search Layer

Search ArXiv, Semantic Scholar, OpenAlex, CrossRef, GitHub, PapersWithCode, Hugging Face, Kaggle, OpenML, Google Dataset Search. Merge and deduplicate evidence.

9. Evidence Store

Store metadata in PostgreSQL and raw artifacts in object storage. Every paper, dataset, and experiment receives immutable provenance.

10. Knowledge Graph

Nodes: Paper, Dataset, Model, Task, Metric, Author, Benchmark, Repository.
Edges: uses_dataset, beats, extends, cites, evaluated_on, implements, trained_on, reproduces.

11. Dataset Intelligence

Score datasets using measurable signals: task match, modality, citations, downloads, maintenance, license, missing values, duplicates, class balance, annotation quality.

12. Dataset Validation

Use Great Expectations, Pandera, Cleanlab, DuckDB and Polars for deterministic cleaning and validation. LLM recommends strategy; libraries execute.

13. Training Planner

Rule engine maps task + hardware + dataset to a recommended architecture and training strategy.

14. Config Generator

Generate YAML/JSON configurations instead of raw code whenever possible to reduce hallucinations and improve reproducibility.

15. Execution Engine

Support PyTorch Lightning, Hugging Face Transformers, TRL, Accelerate, scikit-learn, XGBoost and classical ML.

16. GPU Abstraction

Single interface for Local GPU, RunPod, Modal, Lambda, CoreWeave and Kubernetes clusters.

17. Experiment Manager

Use Optuna or Ray Tune for hyperparameter search. AI proposes hypotheses while optimization frameworks search parameters.

18. Evaluation

Compute benchmark metrics, statistical significance, regression detection, ablations, calibration and robustness reports.

19. AutoResearch Agent

Continuously read new literature, identify promising techniques, schedule experiments, compare against baselines, and automatically create reproducible reports.

20. Multi-Agent Design

Planner Agent
Research Agent
Dataset Agent
Validation Agent
Training Agent
Optimization Agent
Evaluation Agent
Reporting Agent

21. Hallucination Prevention

Search using APIs.
Never invent datasets.
Never clean data with LLMs.
Generate configs over code.
Verify every recommendation with measurable evidence.
Keep provenance for every output.

22. CLI Design

researchforge init
researchforge research
researchforge datasets
researchforge validate
researchforge plan
researchforge train
researchforge tune
researchforge evaluate
researchforge autoresearch
researchforge report
researchforge deploy

23. Suggested Folder Structure

cli/
agents/
planner/
knowledge_graph/
search/
datasets/
training/
evaluation/
experiments/
sdk/
configs/
templates/
docs/

24. MVP Roadmap

Phase 1: Research + Dataset Discovery.
Phase 2: Validation + Planning.
Phase 3: Training + Evaluation.
Phase 4: AutoResearch.
Phase 5: Team collaboration and hosted platform.

25. Business Model

Open-source CLI, hosted cloud execution, enterprise collaboration, managed experiment storage, premium AutoResearch agents, SDK licensing.

26. YC Positioning

ResearchForge becomes the operating system for AI model development—the equivalent of Git + Docker + Vercel for training and evaluating AI models.